

Earnings Debt Prudence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Earnings Debt Prudence (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.39 (0.28), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (15) bps/month with a t-statistic of 2.68 (1.93), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth) is 15 bps/month with a t-statistic of 1.98.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with risk premiums emerging when securities covary with consumption growth or aggregate wealth. While extensive literature documents cross-sectional return predictability through various firm characteristics, the economic mechanisms linking accounting-based signals to systematic consumption risk remain incompletely understood. The interaction between firms' debt management decisions and their earnings quality presents a particularly compelling yet underexplored dimension of this puzzle. Firms exhibiting prudent debt management relative to their earnings capacity may demonstrate differential exposure to aggregate consumption shocks, especially during periods of economic stress when credit constraints bind and precautionary savings motives intensify.

We propose that Earnings Debt Prudence (EDP) captures firms' exposure to long-run consumption risk through multiple channels grounded in consumption-based asset pricing theory. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with high EDP maintain conservative leverage relative to earnings quality, reducing their sensitivity to persistent shocks in aggregate consumption growth. These firms effectively hedge against adverse consumption states when households face binding borrowing constraints, as modeled in [Constantinides \(1990\)](#). The state-dependent nature of EDP's predictive power aligns with the consumption-based explanation: during economic downturns when marginal utility is high, prudently leveraged firms provide insurance against consumption disasters, commanding lower risk premiums consistent with [Barro \(2006\)](#). Furthermore, EDP relates directly to the intertemporal marginal rate of substitution through its connection to firms' financial flexibility. High EDP firms maintain greater capacity to smooth cash flows across states, reducing their covariance with the stochastic discount factor when habit persistence amplifies consumption risk as in [Campbell and Cochrane \(1999\)](#).

This mechanism becomes particularly pronounced during credit crunches when the consumption-wealth channel identified by [Lettau and Ludvigson \(2001\)](#) strengthens, making firms’ debt prudence a first-order determinant of their systematic risk exposure.

We find that a value-weighted long/short trading strategy based on EDP achieves an annualized gross Sharpe ratio of 0.39, with a monthly average abnormal gross return relative to the Fama-French six-factor model of 21 bps/month and a t-statistic of 2.68. The strategy’s performance remains robust across various portfolio construction methodologies, with twenty-five out of twenty-five alphas exhibiting t-statistics exceeding two across different sorting methods and factor models. Among the largest stocks—those with market capitalization greater than the 80th NYSE percentile—the EDP strategy achieves an average return of 24 bps/month with a t-statistic of 2.75, demonstrating that the anomaly is not confined to small, illiquid securities where limits to arbitrage might prevent efficient pricing. The economic significance persists after accounting for transaction costs, with the net average return equaling 15 bps/month and a t-statistic of 2.01 using NYSE quintile sorts with value weighting. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six most closely related strategies from the factor zoo remains significant at 15 bps/month with a t-statistic of 1.98. Conditional double sorts reveal that EDP predicts returns within each size quintile, with alphas ranging from 20 to 28 bps/month among large-cap stocks when controlling for common factor models.

Our study contributes to the intersection of consumption-based asset pricing and accounting-based return predictability by establishing EDP as a robust predictor linked to systematic consumption risk. We extend [Fama and French \(2015\)](#) by demonstrating that debt prudence relative to earnings quality captures risk exposure beyond traditional profitability and investment factors, with our long/short strategy maintaining significant alphas of 21 bps/month after controlling for their

five-factor model plus momentum. Unlike the accruals anomaly documented in [Sloan \(1996\)](#), which primarily reflects mispricing of earnings persistence, EDP’s predictive power stems from its correlation with consumption growth volatility and disaster risk exposure, consistent with rational risk-based explanations. Our findings complement [Hou et al. \(2015\)](#) by showing that while investment-based factors explain substantial cross-sectional variation, the interaction between leverage decisions and earnings quality contains independent information about firms’ exposure to aggregate consumption shocks. Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#), demonstrating that EDP’s performance ranks in the top 19% of gross Sharpe ratios across 212 documented anomalies. The persistence of EDP’s predictive power after transaction costs, with net Sharpe ratios exceeding 89% of existing anomalies, suggests that consumption risk exposure through debt prudence represents a fundamental source of systematic risk. These results have broader implications for understanding how firms’ financial policies determine their covariance with the stochastic discount factor, providing empirical support for consumption-based theories while offering investors a implementable strategy for managing exposure to long-run consumption risks.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms’ debt issuance behavior relative to their operating performance. Earnings Debt Prudence signal requires two key accounting variables.

Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s long-term debt obligations. Earnings before interest and taxes (EBIT) measures the firm’s operating profitability before considering financing costs and tax expenses, providing a clean measure of the earnings generated from core business operations. Both variables are reported in millions of dollars and reflect firms’ actual cash flows and operating performance as disclosed in their financial statements. We construct the Earnings Debt Prudence signal by computing the year-over-year change in long-term debt issuance scaled by lagged earnings before interest and taxes: $(DLTIS(t) - DLTIS(t-1)) / EBIT(t-1)$. This ratio captures the change in a firm’s debt issuance activity relative to its prior-year operating profitability, providing insight into how aggressively firms are expanding their leverage given their earnings capacity. A higher value indicates that a firm is accelerating its debt issuance relative to its earnings base, while a lower or negative value suggests more conservative debt management or debt reduction. We calculate this measure using fiscal year-end values and require non-missing observations for both DLTIS and EBIT across consecutive years. To ensure meaningful scaling, we exclude observations where lagged EBIT equals zero or is negative, as these cases would produce economically uninterpretable ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.58 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal’s interquartile range spans -0.59 to 0.74. Panel B of Figure 1 plots the time-series of

the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 6.19% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.21% per month with a t-statistic of 2.82. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.21% to 0.26% per month and have t-statistics exceeding 2.68 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.16, with a t-statistic of -4.51 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,196 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 11 bps/month with a t-statistics of 2.54. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -15-18bps/month. The lowest return, (-15 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.55. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in

fourteen cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 24 bps/month with a t-statistic of 2.75. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 20 to 28 bps/month with t-statistics between 2.25 and 3.22.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.39 (0.28) is greater than 81% (89%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.23 which ranks the EDP strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$1.23 which ranks the EDP strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 48 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 63 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 210 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.20, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on EDP is 0.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 16-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.08, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.98.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDP is available.

includes EDP grows to \$1225.43.

8 Conclusion

This study provides compelling evidence that Earnings Debt Prudence (EDP) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate economically and statistically significant returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers an annualized gross Sharpe ratio of 0.39, which remains substantial at 0.28 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 21 basis points gross (15 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics of 2.68 and 1.93 respectively, underscores its economic significance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a gross alpha of 15 basis points per month (t-statistic of 1.98) even after controlling for the six factors plus six closely related strategies from the factor zoo, including changes in financial liabilities, net debt financing, and accruals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. EDP appears to capture a distinct dimension of firm financial prudence that is not fully explained by existing factor models or related financial health metrics. This suggests that investors may benefit from incorporating measures of earnings-debt relationships into their security selection processes, particularly given the signal’s ability to generate positive risk-adjusted returns net of implementation costs.

However, several limitations warrant consideration. First, while the net returns remain positive and economically meaningful, the reduction in Sharpe ratio from 0.39 to 0.28 after transaction costs highlights the importance of efficient implementation for real-world applications. Second, the marginal statistical significance of the alpha relative to the comprehensive factor model (t-statistic of 1.98) suggests that while EDP contains unique information, its incremental contribution beyond existing factors may be modest in some specifications.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EDP premium would enhance our understanding of why this signal predicts returns. Does it reflect systematic risk exposure, behavioral biases in how investors process earnings-debt relationships, or limits to arbitrage? Second, examining the signal's performance across different market conditions, particularly during periods of financial stress when debt prudence might be especially valued, could provide insights into its time-varying effectiveness. Third, international evidence would help establish whether the EDP effect is a global phenomenon or specific to certain markets. Finally, combining EDP with machine learning techniques or other signals in a composite strategy might unlock additional predictive power while maintaining economic interpretability.

In conclusion, this paper establishes Earnings Debt Prudence as a robust predictor of cross-sectional stock returns that survives stringent empirical tests and offers practical value to investors. While not revolutionary in magnitude, the signal's persistence after controlling for known factors and transaction costs positions it as a valuable addition to the toolkit of quantitative investors and a worthy subject for continued academic investigation.

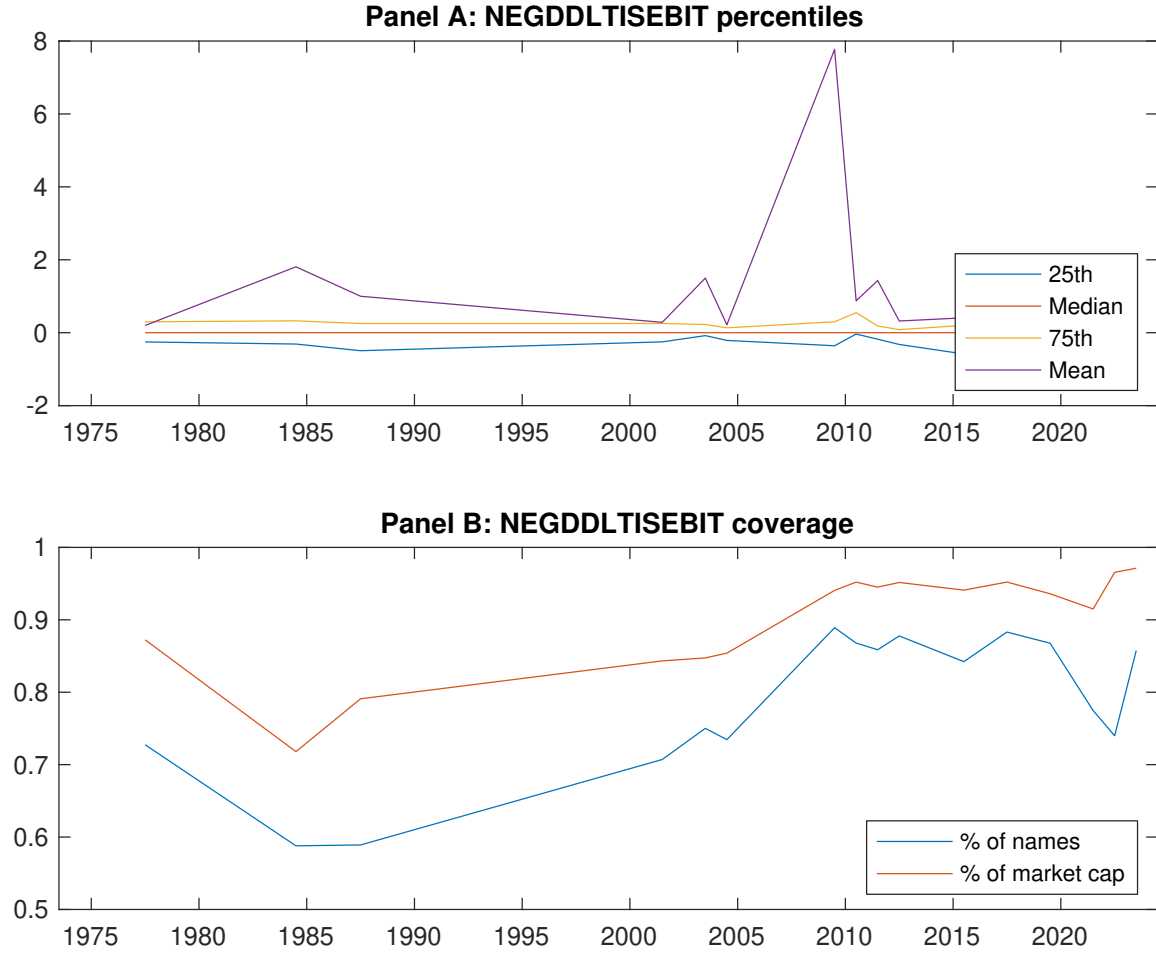


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.57]	0.62 [3.45]	0.66 [3.37]	0.78 [4.43]	0.77 [3.61]	0.21 [2.82]
α_{CAPM}	-0.17 [-2.84]	0.01 [0.25]	-0.00 [-0.04]	0.18 [4.16]	0.05 [0.82]	0.23 [2.99]
α_{FF3}	-0.22 [-3.77]	-0.03 [-0.65]	0.05 [1.04]	0.18 [4.18]	0.04 [0.62]	0.26 [3.41]
α_{FF4}	-0.20 [-3.31]	-0.00 [-0.10]	0.09 [1.78]	0.14 [3.36]	0.03 [0.45]	0.22 [2.93]
α_{FF5}	-0.20 [-3.44]	-0.09 [-2.00]	0.07 [1.24]	0.11 [2.54]	0.03 [0.40]	0.23 [2.98]
α_{FF6}	-0.19 [-3.13]	-0.06 [-1.48]	0.10 [1.79]	0.09 [2.03]	0.02 [0.32]	0.21 [2.68]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	1.10 [79.07]	0.99 [97.73]	0.99 [78.49]	0.96 [97.26]	1.08 [74.41]	-0.01 [-0.73]
β_{SMB}	0.16 [7.85]	-0.11 [-7.19]	-0.04 [-1.93]	-0.04 [-2.55]	0.16 [7.30]	-0.00 [-0.14]
β_{HML}	0.11 [4.19]	0.11 [5.49]	-0.14 [-5.89]	-0.04 [-2.11]	-0.04 [-1.57]	-0.16 [-4.51]
β_{RMW}	0.03 [1.29]	0.13 [6.58]	0.02 [0.95]	0.08 [4.26]	-0.02 [-0.74]	-0.06 [-1.59]
β_{CMA}	-0.10 [-2.37]	0.06 [2.08]	-0.06 [-1.57]	0.15 [5.15]	0.09 [2.22]	0.19 [3.64]
β_{UMD}	-0.03 [-1.90]	-0.04 [-3.61]	-0.05 [-3.81]	0.03 [3.52]	0.01 [0.53]	0.03 [1.90]
Panel C: Average number of firms (n) and market capitalization (me)						
n	647	527	1060	589	635	
me (\$10 ⁶)	1196	3156	2445	3281	1249	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [2.82]	0.23 [2.99]	0.26 [3.41]	0.22 [2.93]	0.23 [2.98]	0.21 [2.68]
Quintile	NYSE	EW	0.11 [2.54]	0.11 [2.70]	0.12 [2.73]	0.12 [2.65]	0.12 [2.73]	0.12 [2.73]
Quintile	Name	VW	0.21 [2.95]	0.24 [3.38]	0.27 [3.77]	0.23 [3.29]	0.21 [3.03]	0.20 [2.77]
Quintile	Cap	VW	0.21 [3.42]	0.24 [3.86]	0.26 [4.18]	0.22 [3.54]	0.19 [3.05]	0.17 [2.68]
Decile	NYSE	VW	0.26 [2.60]	0.25 [2.52]	0.25 [2.50]	0.26 [2.49]	0.21 [2.04]	0.22 [2.10]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.15 [2.01]	0.16 [2.13]	0.19 [2.47]	0.17 [2.24]	0.17 [2.15]	0.15 [1.93]
Quintile	NYSE	EW	-0.15 [-2.55]					
Quintile	Name	VW	0.15 [2.09]	0.18 [2.55]	0.20 [2.85]	0.18 [2.61]	0.16 [2.29]	0.15 [2.08]
Quintile	Cap	VW	0.16 [2.61]	0.19 [3.13]	0.21 [3.37]	0.19 [3.05]	0.16 [2.47]	0.14 [2.20]
Decile	NYSE	VW	0.18 [1.83]	0.17 [1.63]	0.16 [1.60]	0.17 [1.63]	0.12 [1.16]	0.12 [1.17]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.76 [2.82]	0.81 [2.87]	0.90 [3.34]	0.96 [3.25]	0.74 [2.73]	-0.02 [-0.31]	-0.01 [-0.13]	-0.01 [-0.08]	-0.04 [-0.46]	0.04 [0.46]	0.01 [0.15]
	(2)	0.77 [2.91]	0.84 [3.34]	0.82 [3.21]	0.88 [3.50]	0.87 [3.40]	0.10 [1.34]	0.14 [1.73]	0.11 [1.43]	0.12 [1.46]	0.11 [1.39]	0.11 [1.40]
	(3)	0.82 [3.29]	0.83 [3.68]	0.84 [3.49]	0.76 [3.35]	0.88 [3.62]	0.05 [0.71]	0.08 [0.99]	0.09 [1.19]	0.08 [1.05]	0.12 [1.48]	0.11 [1.36]
	(4)	0.67 [2.86]	0.84 [3.93]	0.86 [3.94]	0.76 [3.61]	0.89 [4.02]	0.23 [2.91]	0.25 [3.20]	0.27 [3.41]	0.23 [2.92]	0.30 [3.67]	0.27 [3.28]
	(5)	0.50 [2.51]	0.60 [3.33]	0.61 [3.12]	0.71 [3.90]	0.73 [3.75]	0.24 [2.75]	0.24 [2.81]	0.28 [3.22]	0.25 [2.89]	0.21 [2.39]	0.20 [2.25]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	390	390	391	390	389	37	33	32	33	36	
	(2)	106	106	106	107	106	60	61	60	60	61	
	(3)	75	75	75	75	75	107	109	104	106	107	
	(4)	62	63	63	63	62	226	239	226	237	224	
(5)	58	58	58	58	58	1303	2175	1937	2311	1440		

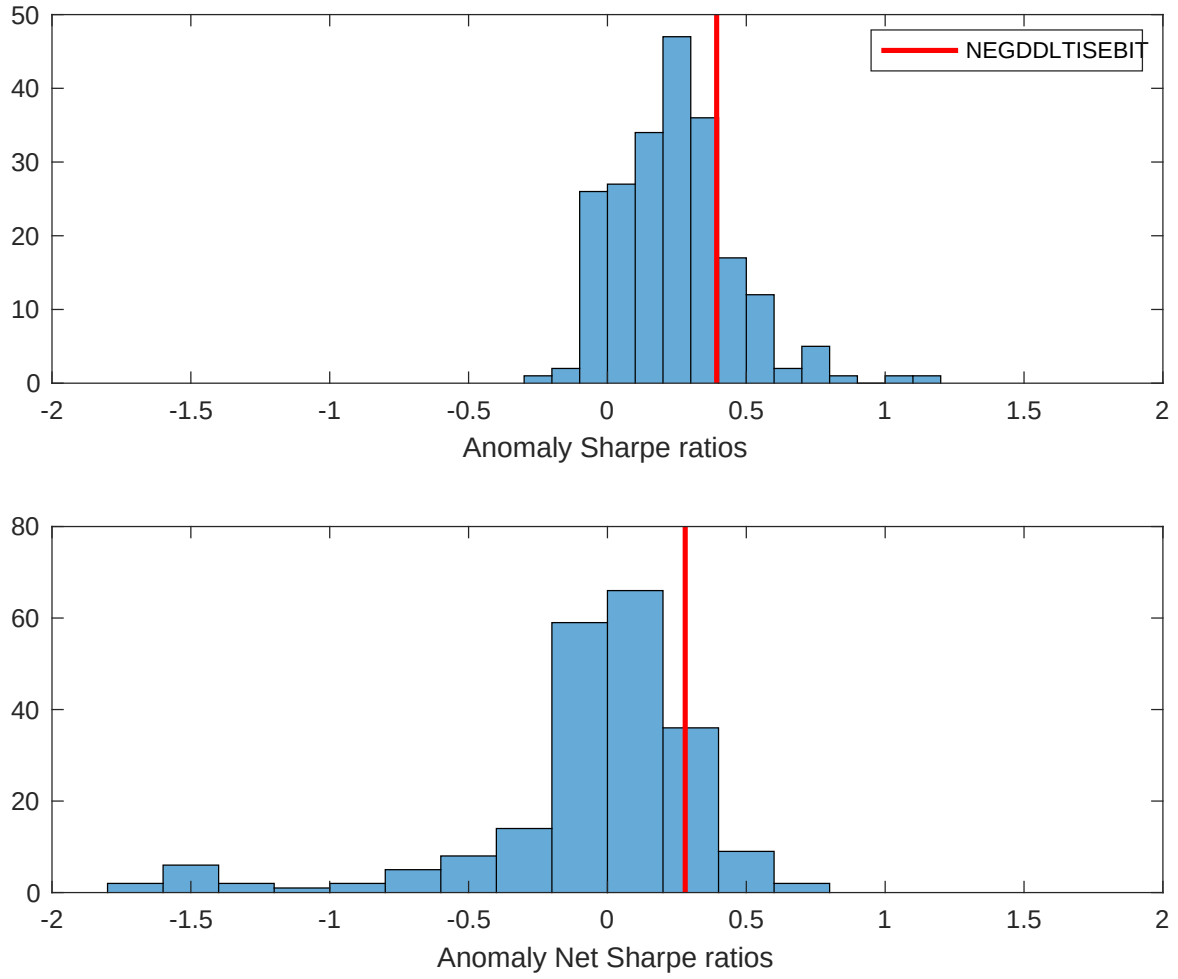


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

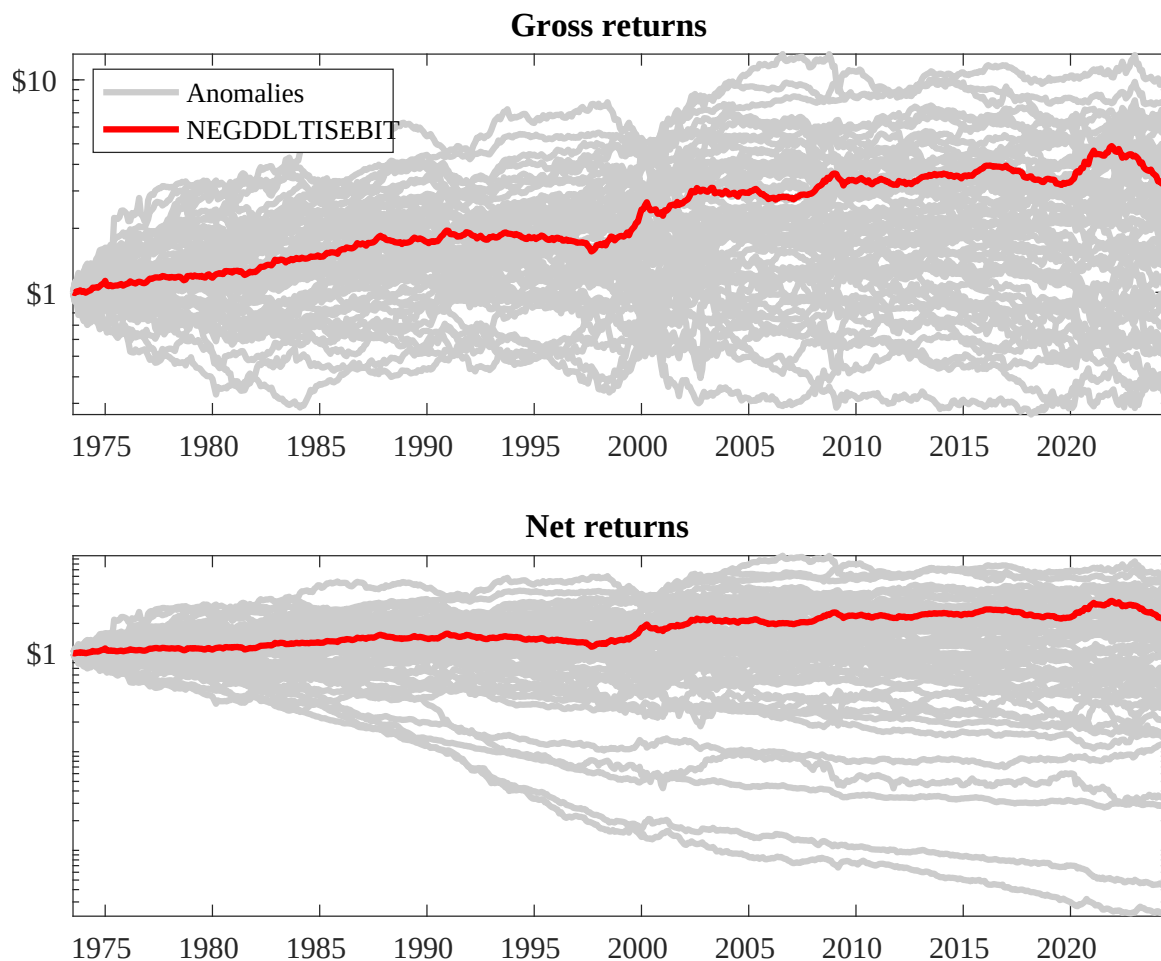
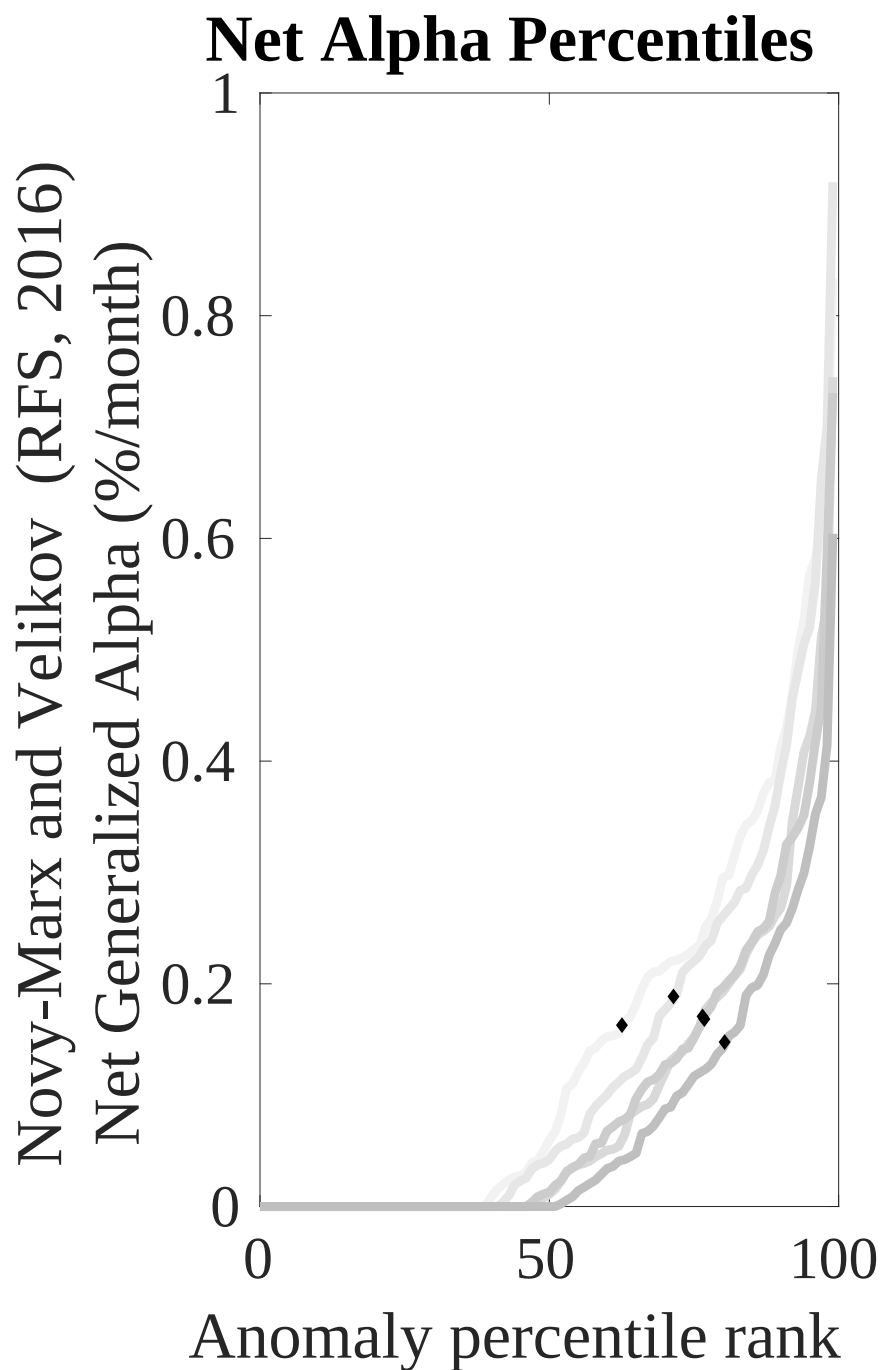
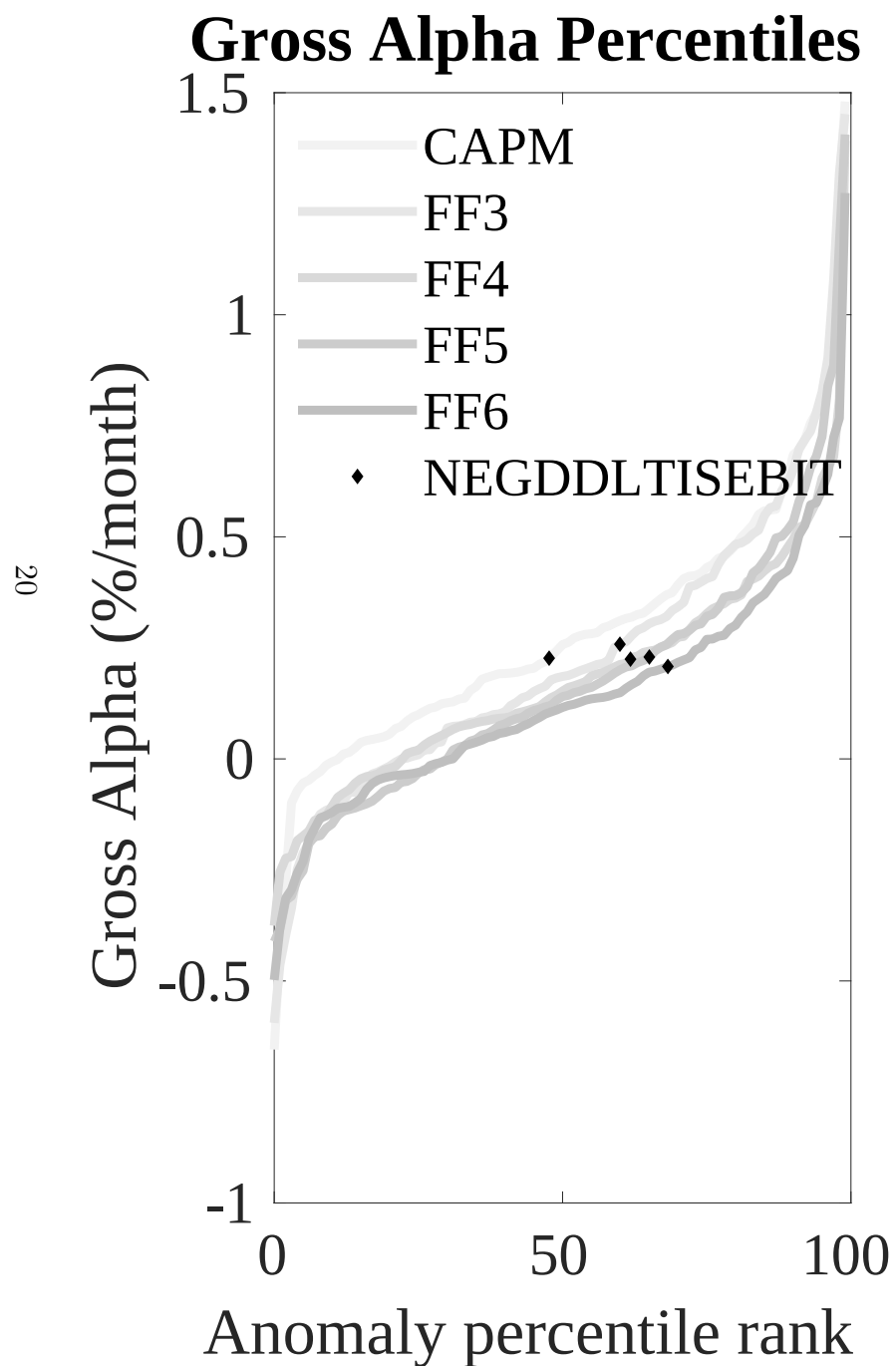


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



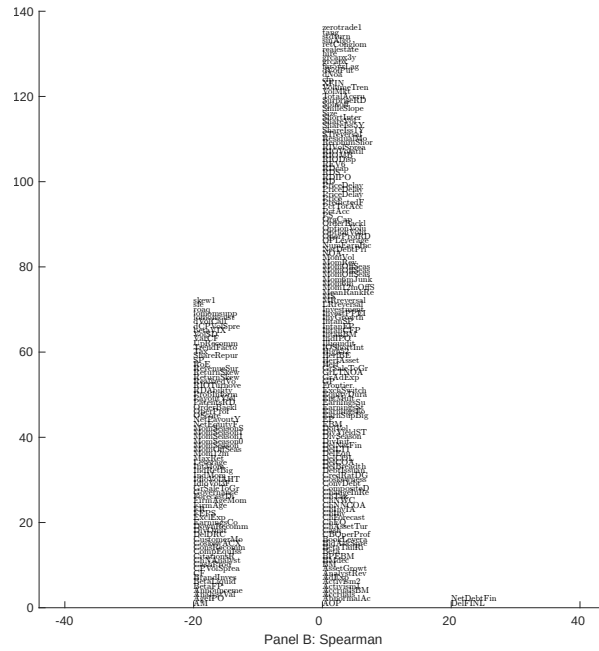
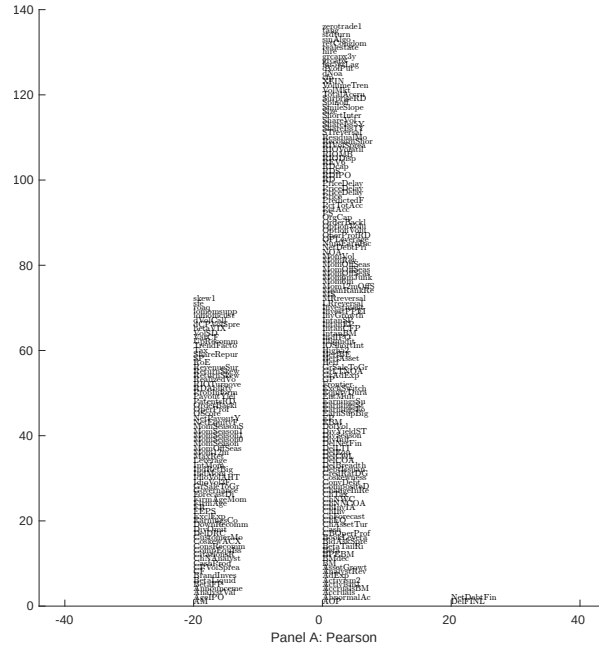


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

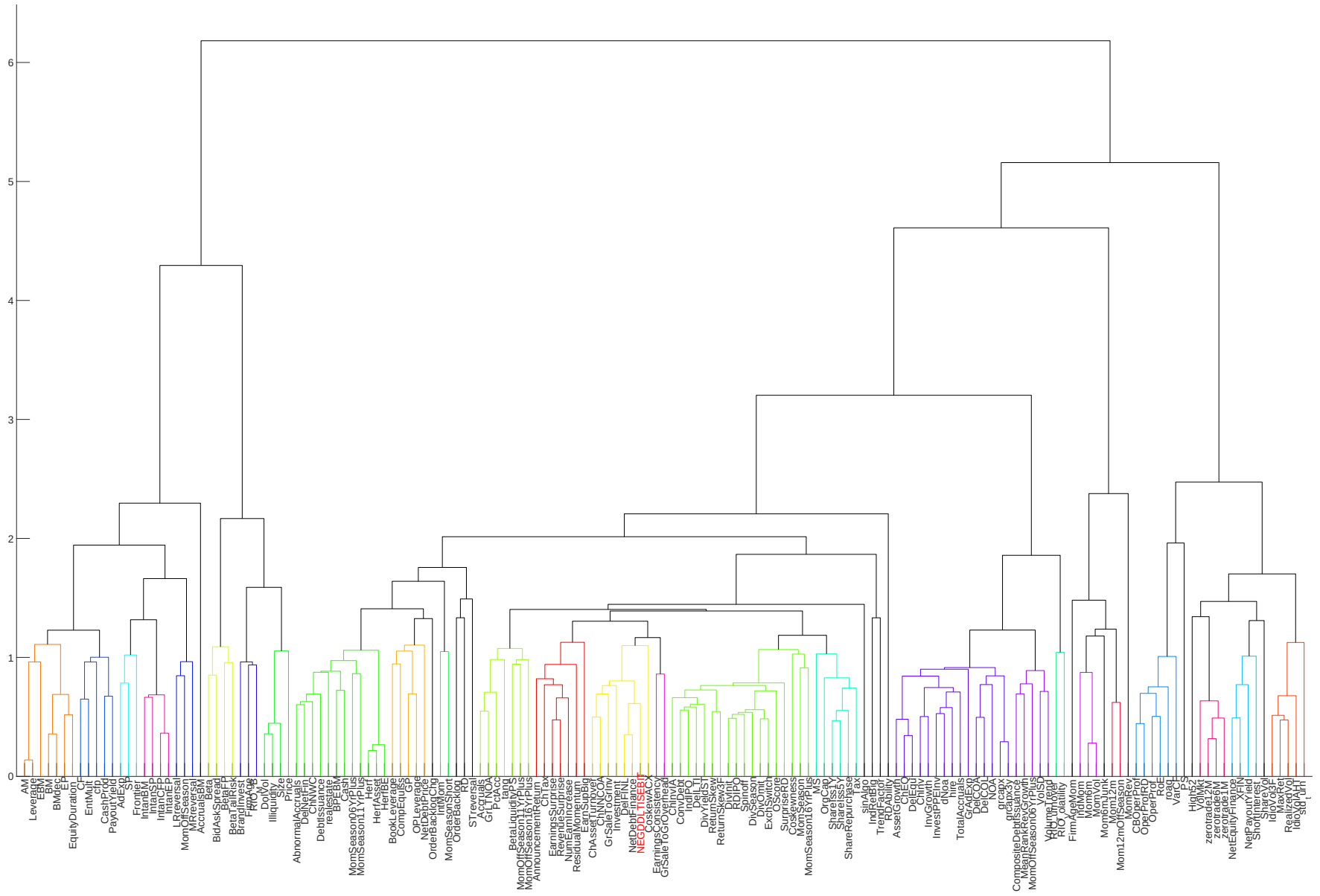


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

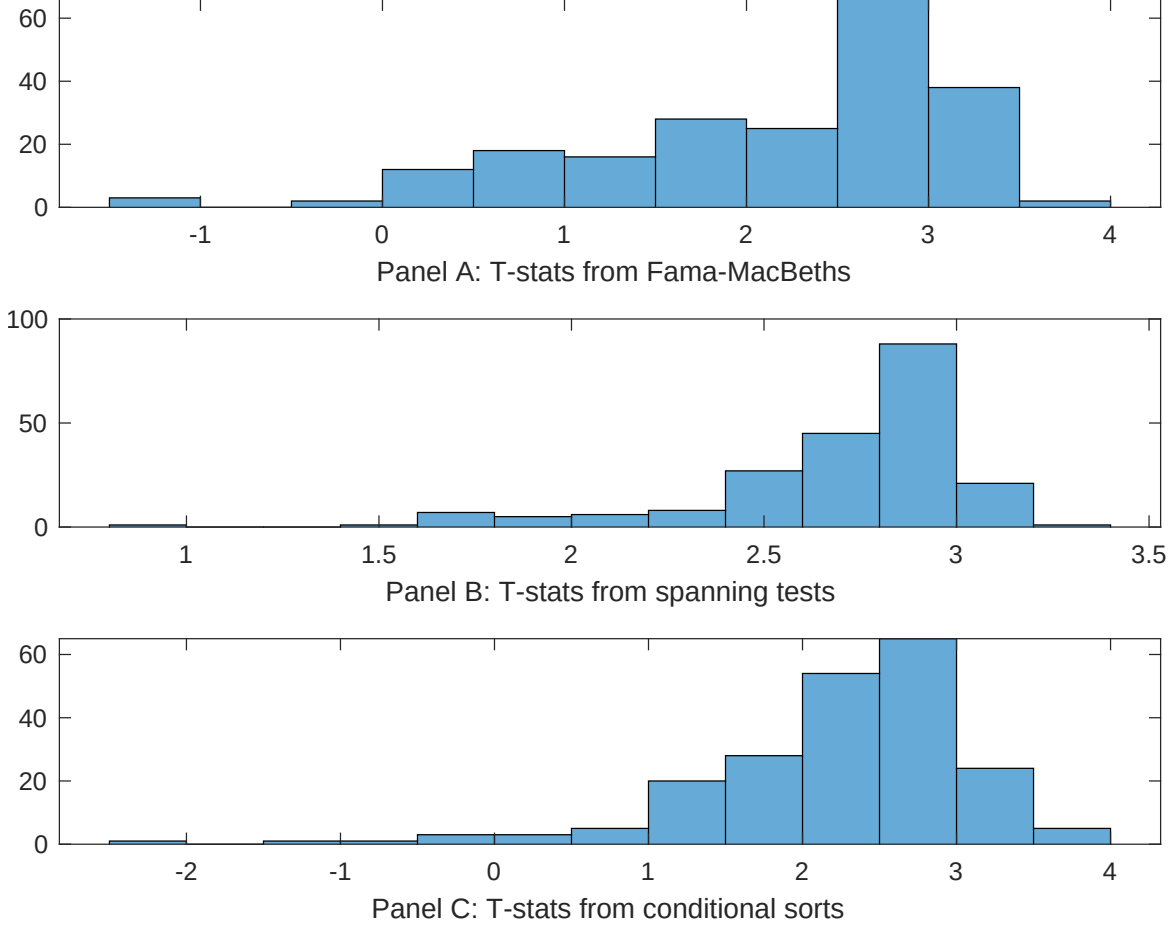


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.34]	0.13 [5.30]	0.13 [5.20]	0.16 [6.29]	0.12 [4.97]	0.13 [5.16]	0.15 [6.03]
EDP	0.77 [0.20]	0.20 [0.50]	0.54 [1.38]	0.11 [2.68]	0.92 [2.34]	0.12 [2.79]	0.19 [0.43]
Anomaly 1	0.18 [9.75]						0.15 [3.08]
Anomaly 2		0.21 [9.43]					0.85 [2.03]
Anomaly 3			0.76 [5.14]				-0.85 [-2.73]
Anomaly 4				0.26 [6.05]			0.20 [4.28]
Anomaly 5					0.14 [4.49]		0.11 [2.84]
Anomaly 6						0.16 [5.16]	0.11 [3.46]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.17 [2.30]	0.18 [2.40]	0.16 [2.08]	0.19 [2.46]	0.18 [2.36]	0.20 [2.57]	0.15 [1.98]
Anomaly 1	24.62 [5.72]						5.35 [0.90]
Anomaly 2		24.97 [6.02]					15.02 [2.79]
Anomaly 3			21.28 [5.92]				11.76 [2.89]
Anomaly 4				11.15 [3.72]			6.47 [2.15]
Anomaly 5					6.10 [1.98]		0.55 [0.18]
Anomaly 6						12.79 [3.86]	7.51 [2.26]
mkt	-0.59 [-0.34]	-1.11 [-0.64]	-0.76 [-0.43]	-1.40 [-0.79]	-0.70 [-0.39]	-1.44 [-0.81]	-1.01 [-0.59]
smb	-2.78 [-1.04]	-2.46 [-0.92]	0.98 [0.37]	-2.32 [-0.85]	0.91 [0.33]	-0.79 [-0.29]	-2.87 [-1.03]
hml	-13.88 [-4.11]	-14.50 [-4.31]	-15.88 [-4.72]	-14.74 [-4.30]	-13.94 [-3.95]	-15.07 [-4.41]	-13.98 [-4.12]
rmw	-7.85 [-2.28]	-8.38 [-2.43]	-3.08 [-0.90]	-4.69 [-1.35]	-3.63 [-0.99]	-6.99 [-2.00]	-6.43 [-1.78]
cma	11.42 [2.15]	13.18 [2.53]	25.09 [4.83]	18.60 [3.57]	17.67 [3.31]	18.76 [3.61]	15.72 [2.82]
umd	1.33 [0.75]	1.86 [1.06]	3.10 [1.79]	2.04 [1.13]	3.01 [1.68]	1.99 [1.10]	0.08 [0.04]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	10	10	10	7	6	7	14

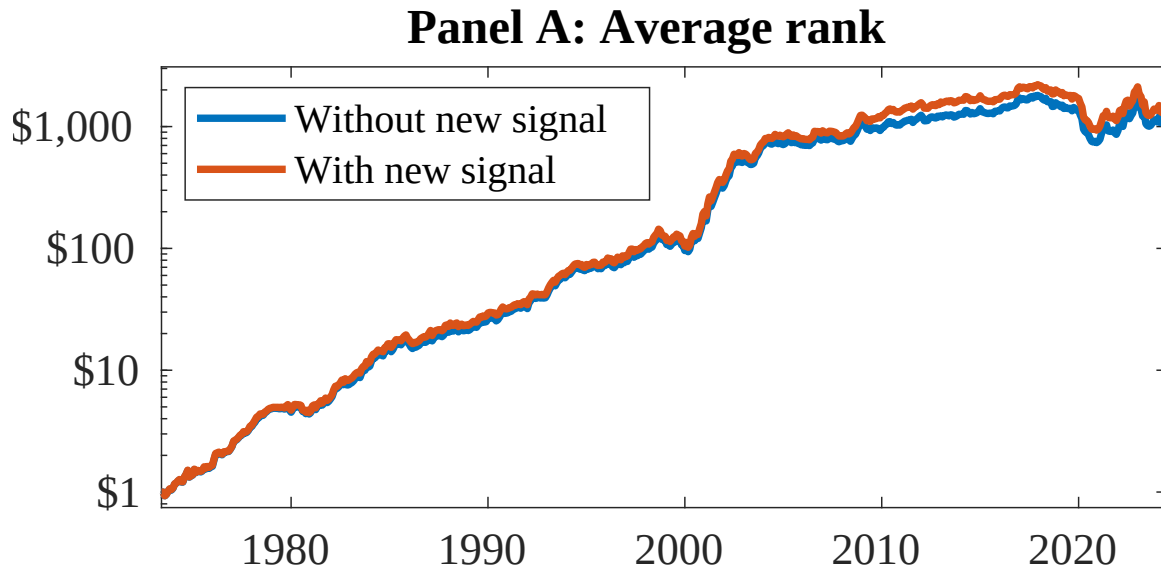


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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